HIGH ALTITUDE MEDICINE & BIOLOGY
Volume 27, Number 1, 2026
Mary Ann Liebert, A Part of Sage
DOI: 10.1177/15578682251365931

Review

Pharmacological Prophylaxis and Supplemental Oxygen for Unacclimatized Rescuers at Very High Altitude: Scoping Review and 2025 Joint Recommendations of the International Commission for Mountain Emergency Medicine and the International Society for Mountain Medicine

Kyle McLaughlin,[1–3] Steve Roy,[2–4] Marika Falla,[3,5] Giacomo Strapazzon,[2,3,5] Andrew M. Luks,[3,6] Ken Zafren,[2,3,7] Hermann Brugger,[2,3,5] Martin Musi,[2,8] Iztok Tomazin,[2,9] John Ellerton,[2,3] Ghan Bahadur Thapa,[10,11] and Peter Paal[2,3,12]

Abstract

McLaughlin, Kyle, Steve Roy, Marika Falla, Giacomo Strapazzon, Andrew M. Luks, Ken Zafren, Hermann Brugger, Martin Musi, Iztok Tomazin, John Ellerton, Ghan Bahadur Thapa, and Peter Paal. Pharmacological prophylaxis and supplemental oxygen for unacclimatized rescuers at very high altitude: scoping review and 2025 joint recommendations of the International Commission for Mountain Emergency Medicine and the International Society for Mountain Medicine. *High Alt Med Biol.* 27:60–77, 2026.

Background: Mountain rescuers and pilots rapidly ascending to altitudes above 3,500 m are exposed to the detrimental effects of hypobaric hypoxia, including cognitive and physical impairment, as well as high-altitude illness (HAI). We conducted a scoping review of oxygen supplementation and pharmacologic measures to improve cognitive and physical performance and prevent HAI in unacclimatized rescuers rapidly ascending above 3,500 m during rescue missions.

Methods: Following Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, 723 articles were screened, 133 reviewed and a total of 50 articles were included for data extraction, based on the intervention: 6 on oxygen, 29 on acetazolamide (AZ), 17 on dexamethasone (DEX), 3 on nifedipine, and 5 on phosphodiesterase-5 inhibitors.

Discussion: Supplemental oxygen improves physical and cognitive performance at high altitude and is recommended for rapid ascent rescues >30 minutes between 3,500 and 4,000 m, and for rescues of any duration above 4,000 m. If oxygen is administered, pharmacological prophylaxis is not required. If oxygen is unavailable, AZ or DEX can be used for rapid ascent rescues above 3,500 m for longer than 3 hours to reduce the incidence and severity of acute mountain sickness. At altitudes above 5,000 m or for rescues requiring prolonged physical work, the use of both AZ and DEX is recommended.

[1]Department of Emergency Medicine, Canmore General Hospital; Clinical Lecturer, Cumming School of Medicine, University of Calgary, Canada.
[2]International Commission for Mountain Emergency Medicine (ICAR MedCom), Zurich, Switzerland.
[3]International Society for Mountain Medicine, Basel, Switzerland.
[4]Department of Critical Care Medicine, Cumming School of Medicine, University of Calgary, Canada.
[5]Institute of Mountain Emergency Medicine, Eurac Research, Bolzano, Italy.
[6]Division of Pulmonary, Critical Care and Sleep Medicine, University of Washington. Seattle, WA, USA.
[7]Department of Emergency Medicine, Stanford University Medical Center, Palo Alto, CA, USA.
[8]Associate Professor of Emergency Medicine, University of Colorado School of Medicine, Aurora, CO, USA.
[9]Mountain Rescue Association of Slovenia, Department of Family Medicine, Faculty of Medicine, University of Ljubljana, Slovenia.
[10]Department of Emergency Medicine and Prehospital Care, Grande International Hospital, CIWEC Hospital, Kathmandu, Nepal.
[11]Mountain Medicine Society of Nepal, Kathmandu, Nepal.
[12]Department of Anaesthesiology and Intensive Care Medicine, St. John of God Hospital, Paracelsus Medical University, Salzburg, Austria.

© The Authors 2026. Published by Mary Ann Liebert (NY), LLC. This article is distributed under the terms of the Creative Commons Attribution 4.0 License (https://creativecommons.org/licenses/by/4.0/) which permits any use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (https://us.sagepub.com/en-us/nam/open-access-at-sage).

Conclusions: To enhance the safety and effectiveness of high-altitude rescues, we provide recommendations for the use of supplemental oxygen and pharmacologic prophylaxis to reduce the risk of HAI and improve cognitive and physical performance during rapid ascents to altitudes >3,500 m.

Keywords: acetazolamide; dexamethasone; high altitude; high-altitude illness; mountain rescue; nifedipine; oxygen

Introduction

Given the large number of people traveling to high altitudes (HAs) annually (Burtscher et al., 2001) and the increasing number of climbers aiming to summit the world's highest mountains (Global Rescue, 2024; Salisbury et al., 2021), the risk of accidents and the need for mountain rescue operations are rising (Gasser, 2022).

Altitudes ranging from 1,500 to 3,500 m are considered "HA," 3,500 to 5,500 m as "very high altitude" (VHA), and above 5.500 m as "extreme altitude" (Roach et al., 2016). In this review, "VHA" refers to the altitudes greater than 3,500 m. Rescue operations at VHA present unique challenges beyond typical low-elevation mountain rescue risks. (Milani et al., 2023; Wu, 2011). These rescues often involve rapid ascent, usually by helicopter. They can range from quick patient transports to complex rescues that require prolonged physical ground work in complicated terrain and challenging weather. The modes of travel and patient conditions vary.

Acute exposure to hypobaric hypoxia at VHA can impair cognitive and physical function and increase the risk of developing high-altitude illnesses (HAI) such as acute mountain sickness (AMS), high-altitude cerebral edema (HACE), and high-altitude pulmonary edema (HAPE) (Burtscher et al., 2021; Castellani et al., 2021; Fulco et al., 1998; Grocott et al., 2009; Mallet et al., 2023; Paal et al., 2022; Willmann, 2015; Yan, 2014). The risk of HAI varies depending on several factors, including genetic predisposition, length of time spent at HA, maximum altitude reached, ascent rate, flying to HA, and sleep elevation (Burtscher et al., 2023; Hackett et al., 1976; Hansen et al., 1967; Hillenbrand et al., 2006; Poudel et al., 2025). Cognitive and psychiatric symptoms should also be considered, as neuropsychiatric symptoms can be more subtle than the classical symptoms of HAI (Falla et al., 2024a).

For the general population traveling to HAs, a gradual ascent with time for acclimatization is the most effective way to reduce the risk of HAI (Luks et al., 2024). However, mountain rescue operations do not allow time for a gradual ascent, and many rescuers may not have spent sufficient time at HA prior to a mission to benefit from residual acclimatization effects. Therefore, other measures, including supplemental oxygen and pharmacologic prophylaxis against HAI, must be considered to mitigate the risks of hypobaric hypoxia and ensure safe performance during rescue.

Most published guidelines on HAI prevention using pharmacologic prophylaxis and other measures focus on the general public's standard ascent profiles (Luks et al., 2024) and do not address the specific challenges of rapid ascent to VHA during rescue missions. Furthermore, these guidelines fail to consider the differences between air and ground rescue or the distinct needs of pilots and rescuers. Protocols for VHA rescues have been established in certain regions, but these vary greatly depending on the altitude, topography, and local political and economic factors (Civil Aviation Authroity of Nepal CAAN, 2013; European Union Aviation Safety Agency EASA, 2007; Federal Aviation Administration, 2021; Government of Canada, 2023; Ollig, 2024; Smith, 2020).

The objective of this scoping review is to address these gaps in knowledge and evaluate the literature on oxygen supplementation and pharmacologic prophylaxis to improve cognitive and physical performance and prevent HAI in unacclimatized rescuers stationed at low altitudes who are swiftly deployed to VHA. The findings will inform recommendations for mountaineers, rescuers, pilots, and rescue organizations to improve the safety of rescue missions above 3,500 m.

Methods

We adhered to the Preferred Reporting Items for Scoping Reviews and Meta-Analysis Protocols (Tricco et al., 2018). Objectives, inclusion and exclusion criteria, and Population-Intervention-Comparator-Outcome (PICO) questions were defined.

A systematic literature search was conducted in PubMed, Cochrane Database, and Embase through Jun 24, 2024. We sequentially evaluated the titles, abstracts, and full texts of all publications identified through our search for potentially relevant publications. A manual search of the eligible studies followed a database search.

Two authors identified studies that utilized supplemental oxygen during rapid ascent to VHA using the search string outlined in Supplementary Data S1. The search strategy was organized according to the Population–Concept–Context mnemonic, as recommended for scoping reviews by the JBI Collaboration (Peters et al., 2020). Data were extracted from the final six studies (Table 1).

Two authors, along with an experienced librarian, developed search strings to identify articles on pharmacological prophylaxis for HAI during rapid ascent (>300 m/day) to VHA (Supplementary Data S2). All publications were independently reviewed and disagreements were resolved by a third author. The final search results were exported to the Covidence software and duplicates were removed. Data from the final 44 studies were evaluated for quality and relevance using the PICO questions (Tables 2–5).

Recommendations were developed and graded by consensus based on the strength and quality of evidence using the system of the American College of Chest Physicians (Guyatt et al., 2006). Finally, the article was discussed and accepted by the International Commission for Alpine Rescue Medical commission at the ICAR Congress 2024 in Thessaloniki, Greece and endorsed by the International Society for Mountain Medicine in January 2025.

Results

We screened 723 articles, reviewed the full text of 133 articles, and included 50 studies for data extraction. The

Table 1. Studies of Supplemental Oxygen for Rapid Ascent Above 3,500 m

Study	*Study design, N*	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Crowley et al., 1992)	RCT, 13	35% O_2 treatment group vs. 20% O_2 control group	Simulated altitude in hypobaric chamber	Healthy, non-acclimatized military males	4,300	4,300	4,300	Cognitive Performance: O_2 improved code substitution ($p = 0.04$), serial addition/subtraction ($p = 0.03$) and activity index in mood ($p = 0.04$) on day 1 only. No significant effects seen past day 1.
(Gerard et al., 2000)	RCT, 24	18% oxygen in the test room was called the "ambient air 5,000 m treatment," and 23% oxygen was designated the "6% oxygen-enriched 5,000 m treatment"	Ground vehicle	Healthy non-acclimatized adults	3,800 then simulated to 5,000 m	Barcroft facility at 3,800 m	3,800	Cognitive Performance: 6% O_2 enrichment resulted in higher arterial oxygen saturations (93.0 vs. 81.6%), quicker reaction times, improved hand-eye coordination, and more positive sense of well-being ($p < 0.05$)
(Silva-Urra et al., 2011)	RCT, 23	Walk 1: 2 km walk at sea level; Walk 2: No oxygen at 5,050 m; Walk 3: Oxygen 1 l/min vs. 3 l/min at 5,050 m.	Ground vehicle	Healthy non-acclimatized students	5,050	5,050	5,050	AMS Severity: O_2 improved AMS severity score. Only men were statistically significant. O_2 at 3 l/min >1 l/min at reducing AMS symptoms. Exercise capacity: Improved walk times with O_2. Only 3 l/min had lower perceived exertion vs. no O_2 ($p < 0.05$) but performance at altitude was still poorer then at sea level despite the O_2.
(Legg et al., 2016)	Interventional, cross over, 36	100% O_2	Simulated altitude using hypobaric chamber	Healthy non-acclimatized Air Force personnel	3,657	3,657	3,657	Cognitive performance: Increased fatigue and reduced vigor at 3,657 m restored by O_2 100%; no clear effect on logical reasoning.
(Falla et al., 2024a)	RCT, 36	O_2, 2 l/min (via OxyMask)	Simulated altitude using hypobaric chamber	Healthy non-acclimatized volunteers from EMS and search and rescue (SAR) services	3,800	4,000	3,800	Cognitive performance: Reduced sustained attention and reaction time that improved with O_2 supplementation
(Beer et al., 2024)	Interventional, cross over, 12	21% vs. 100% O_2 therapy in both gradual and rapid decompression	Simulated altitude using hypobaric chamber	Air Force personnel	4,877	7,620	4,877	Cognitive performance: Declined during ascent with 21% O_2, with key metrics, including composite score, falling further (−75% vs. −50%) after rapid decompression. 100% O_2 prevented cognitive impairment in both gradual and rapid decompression.

AMS, acute mountain sickness; RCT, randomized controlled study.

TABLE 2. STUDIES OF ACETAZOLAMIDE FOR RAPID ASCENT ABOVE 3,500 M

Study	*Study design, N*	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Hackett et al., 1976)	RCT, 278	AZ 250 mg BID × 4 days	Trekking	Adult trekkers in Himalayas many with previous altitude exposure	Variable	Variable	350–400	AMS Incidence: No significant difference AZ vs. placebo for those hiking from Kathmandu. AZ 14% vs. 39% placebo for those who flew to Lukla (2,800 m) ($p < 0.01$)
(Greene et al., 1981)	RCT, 24	Group 1: AZ 500 mg OD SR 5 days before and during 1st ascent, then placebo (same protocol) for 2nd ascent. Group 2: Reverse order.	Trekking	Non-acclimatized, lowlander British adults in Africa	3,895	5,896	1,035–1,475	AMS Severity: AZ 4.8 vs. placebo 14.3 ($p < 0.05$)
(Larson et al., 1982)	RCT, 64	AZ 250 mg TID vs. placebo 1 day before ascent	Ground vehicle, trekking	Healthy, sea level climbers in Cascade Range	4,394	4,394	2,197	AMS Incidence: AZ 17.2% vs. 66.7% placebo ($p < 0.001$); Crossover (climbed Rainier 1 year later) 4 had AMS on placebo and 0 on AZ. 83% said they felt better on AZ
(White, 1984)	RCT, 11	AZ 500 mg SR or placebo 2 days prior to ascent and continued for 5 days.	Ground vehicle, trekking	Healthy non-acclimatized students in the Andes	3,600	3,600	2,400	AMS Severity (ESQ): AZ 32 vs. 62 placebo ($p = 0.055$)
(McIntosh and Prescott, 1986)	Observational study, 48	AZ 500 mg OD 1 day before ascent vs. control group not taking meds	Trekking	Healthy, non-acclimatized adults on Kilimanjaro	4,065	5,895	1,016	AMS Severity: AZ had significantly lower AMS scores vs. controls ($p < 0.02$)
(Ellsworth et al., 1987)	RCT, 47	AZ 250 mg, DEX 4 mg, or placebo TID 1 day before ascent.	Ground vehicle, trekking	Healthy, non-acclimatized lowlander adults on Mt Rainier	4,392	4,392	2,196	AMS Incidence: AZ 53.3% vs. 76.9% placebo ($p > 0.05$)
(Zell and Goodman, 1988)	RCT, 32	AZ 250 mg BID, DEX 4 mg QID, both or placebo 2 days before ascent and continued for 2 days at altitude.	Ground vehicle, trekking	Healthy, non-acclimatized adults.	2,710	4,050	1,355	AMS incidence: AZ 16% vs. placebo 38% ($p = 0.04$); AZ + DEX 6% vs. placebo 38% ($p = 0.004$)
(Ellsworth et al., 1991)	RCT, 18	AZ 250 mg TID or DEX 4 mg TID vs. placebo 1 day prior to ascent	Ground vehicle, trekking	Healthy, non-acclimatized lowlander adults on Mt Rainier	4,392	4,392	2,196	AMS incidence: significantly lower in DEX vs. placebo or AZ groups; AMS Severity (ESQ): symptom scores for DEX (0.26 and 0.20) were significantly lower than AZ (0.80 and 1.20; $P = .025$) and placebo (1.11 and 1.45; $P = .025$).
(Bernhard et al., 1998)	RCT, 13	AZ 500 mg OD + DEX 4 mg BID vs. AZ 500 mg OD + placebo	Airplane, ground vehicle	Healthy, non-acclimatized lowlander adults in Andes. 50% had previously experienced AMS.	1,645	5,334	1,645	AMS Incidence: No significant difference at 3,689 m day 1 and 2; AZ + DEX 0–17% vs. 86–100% AZ + placebo at 5,334 m on Day 3; AZ + DEX 17–50% vs. 43–86% AZ + placebo at 5,334 m on Day 4.
(Hussain et al., 2001)	RCT, 24	AZ 250 mg BID vs. DEX 4 mg BID vs. AZ 250 mg +DEX 4 mg BID vs. placebo 1 day before ascent	Ground vehicle, trekking	Healthy, non-acclimatized lowlander adults in Karakoram range	4,063	4,578	4,063	AMS severity (ESQ): Lower score (5.3) for AZ+ DEX ($p < 0.01$) compared to placebo (11.8), Az alone (16.8) or DEX alone (14.5)
(Basnyat et al., 2003)	RCT, 155	AZ 125 mg BID	Trekking	Healthy non-Nepali trekking in the Himalayas.	2,137–2,937	4,937	497–683	AMS incidence: AZ RR 50.6% (NNT 8); AMS Severity (LLS): Severe AMS 0% in AZ vs. 30% in placebo ($p = 0.014$)
(Hussain and Aslam, 2003)	RCT, 44	AZ 250 mg BID vs. DEX 4 mg BID vs. combination of both vs. placebo 1 day before and during ascent.	Ground vehicle, trekking	Healthy non- acclimatized lowlander adults in Karakoram range	4,063	4,578	4,063	Compared to placebo and DEX subgroups, AZ significantly increased RR and PaO_2, while decreasing PCO_2 and pH thus expediting acclimatization.

(*continued*)

TABLE 2. (CONTINUED)

Study	Study design, N	Intervention	Mode of transport	Population	Delta elevation (m)	Max elevation (m)	Ascent rate (m/day)	Results
(Carlsten et al., 2004)	RCT, 32	AZ 125 mg BID vs. AZ 250 mg BID vs. placebo BID	Airplane	Minimally exercising lowlander adult vacationers	3,630	3,630	3,630	AMS Severity (LLS): Only 250 mg BID had a significant reduction in AMS scores at 24 hours (2.7 to 0.6; $p = 0.008$)
(Chow et al., 2005)	RCT, 57	Ginkgo Biloba 120 mg OD 5 days prior to ascent vs. AZ 250 mg BID started on day of ascent vs. placebo	Ground vehicle	Non-acclimatized lowlander adults in California	2,570	3,800	2,570	AMS Incidence: AZ 30% vs. 60% placebo vs. 65% Ginkgo Biloba ($p = 0.06$)
(Basnyat et al., 2006)	RCT, 204	AZ 125 mg BID vs. 375 mg BID vs. placebo	Trekking	Healthy non-Nepali trekking in the Himalayas.	1,488	4,928	336–354	AMS Incidence: No difference 250 mg (24%) vs. 750 mg (21%). Both were superior to placebo (51%). AMS Severity: No statistical difference in all 3 groups.
(Hillenbrand et al., 2006)	RCT, 109	AZ 250 mg OD vs. placebo x 7d	Trekking	Adult male Nepali trekking porters-28 were Highlanders (born >3,050 m) and 81 were lowlanders. Not controlled for acclimatization.	1,470	4,930	43% were >300 m/day	AMS Incidence: AZ 6.4% vs. 5.5% placebo. All +AMS were lowlanders; AMS increased based on rate of ascent in lowlanders: >300 m/day 28.5% vs. <300 m/day 6.5% ($p = 0.013$)
(van Patot et al., 2008)	RCT, 44	AZ 125 mg BID vs. placebo 3 days before ascent and for 1 day at altitude	Ground vehicle	Healthy, adults who reside at 1,400–1,600 m	2,700	4,300	2,700	AMS Incidence: AZ 14% vs. placebo 45% (NNT = 3; $p = 0.02$); AMS Severity: AZ 9% vs. 55% placebo for severe AMS (NNT = 3; $p = 0.007$)
(Subudhi et al., 2011)	RCT, 20	AZ 250 mg, DEX 4 mg or placebo TID 1 day before ascent in chamber and continued at altitude	Hypobaric chamber	Healthy adults who resided at 1650 m × 1 year	3,225	4,875	3,225	AMS Severity (LLS): AZ 1.3 vs. DEX 1.3 vs. 4.0 placebo ($p < 0.05$)
(Ke et al., 2013)	RCT, 28	Group A: AZ 250 mg SR OD + DEX 4 mg BID; Group B: AZ 250 mg SR OD + placebo 1 day before ascent and during next 4 days at altitude.	Airplane	Healthy, non-acclimatized lowlander adults in Tibet	3,261	3,658	3,261	AMS Severity (LLS): AZ 1 vs. Gingko 5 vs. placebo 3 ($p = 0.016$)
(Wang et al., 2013)	RCT, 21	AZ 125 mg BID vs. placebo 3 days before and during 1 day at altitude.	Airplane	Healthy, non-acclimatized lowlander adults in Asia	3,159	3,561	3,159	AMS Severity (LLS): AZ 1.3 vs. 3.1 placebo ($p < 0.05$)
(Eigenberger et al., 2014)	Case series, 916	Variable	Trekking	Adults on commercial trekking companies climbing Kilimanjaro	3,800–4,100	5,895	542–820	AMS Incidence: AZ 72% vs. 70% placebo
(Bradwell et al., 2014)	RCT, 20	AZ 250 mg BID vs. placebo 3 days before ascent	Cable car	Healthy, non-acclimatized adults in Alps	1,409–1,851	3,459	1,409–1,851	AMS Severity (LLS): In 10 matched pairs of subjects, 70% had higher AMS scores in placebo compared to their AZ partner. Highest AMS scores were in 4 of the placebo individuals with the lowest resting SpO_2 ($p < 0.05$)
(Burtscher et al., 2014)	RCT, 155	AZ 250 mg or placebo taken 10 hours and 1 hour before ascent	Ground vehicle, cable car	Healthy, non-acclimatized adults in Alps	2,880	3,480	2,880	AMS Incidence: AZ 25% vs. 71% placebo ($p = 0.07$)

(*continued*)

TABLE 2. (CONTINUED)

Study	*Study design, N*	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Sikri et al., 2019)	RCT, 1302	Group A: AZ 250 mg SR OD + DEX 4 mg BID; Group B: AZ 250 mg SR OD + placebo 1 day before ascent and during 4 days at altitude. Controls had no medication.	Ground vehicle	Healthy Indian Army male soldiers from lowland ethnicity	3,505	3,505	875	AMS Incidence: AZ + DEX 1.56% vs. 1.59% AZ + placebo vs. 19.14% in controls ($p < 0.0001$); AMS Severity (LLS 10–12): AZ + DEX 0% vs. 0% AZ + placebo vs. 4.8% in controls had severe AMS; Exercise capacity: AZ + DEX decreased by 7.83% vs. 12.97% in AZ + placebo
(Lipman et al., 2019)	RCT, 104	AZ 125 mg vs. placebo for 1 dose the night before, then all had AZ 125 mg BID for the 1 day trek	Ground vehicle, trekking	Healthy, non-acclimatized, lowlander adults in Sierra Nevada range	2,305	3,545	2,305	AMS incidence: night before 39% vs. day of ascent 48% ($p = 0.46$); NNT: Night before 3.7 vs. Day of 5.6; AMS Severity (LLS): night before 22% vs. day of ascent 10% for severe AMS (95% CI −28 to 3.6);
(McIntosh et al., 2019)	RCT, 73	AZ 62.5 mg vs. 125 mg BID 1 day before flight to altitude	Airplane, trekking	Healthy adults planning on trekking to Everest base camp	3,620–4,115	5,020–5,515	488–524	AMS incidence: 55.3% (Reduced-dose group) vs. 60% (standard dose group) ($p = 0.47$); AMS Severity (LLS 10–12): 2.9% (Reduced-dose group) vs. 1.7% (standard dose group) had severe AMS ($p < 0.05$)
(Bradbury et al., 2020)	RCT, 10	AZ 250 mg BID vs. placebo 2 days prior to ascent	Hypobaric chamber	Healthy non-acclimatized military men	3,500	3,500	3,500	AMS incidence: AZ 0% vs. 40% placebo. Exercise capacity: 2-mile treadmill AZ 20.7 minutes vs. 22.7 minutes placebo; repeat 2 mile after 24 hr at altitude: AZ 21.5 vs. 21.1 minutes placebo.
(Berger et al., 2022)	RCT, 18	AZ 250 mg TID vs. placebo 2 days before ascent and during 4 days at altitude	Trekking, cable car, climb	Healthy, non-acclimatized adults in Alps	3,429	4,559	3,429	AMS incidence: 57% vs. 83% placebo ($p = 0.31$); AMS Severity (LLS): AZ 4.23 vs. placebo 7.29 ($p = 0.01$); HAPE Incidence: AZ 43% vs. 67% placebo ($p = 0.39$)
(George et al., 2023)	RCT, 120	AZ 250 mg vs. NIF 30 mg SR vs. TDFL 10 mg vs. placebo BID 1 day before ascent and during 3 days at altitude.	Airplane	Military HAPE-susceptible lowlander adults in India who were reintroduced to high altitude 4 weeks after full recovery from HAPE.	3,500	3,500	3,500	HAPE incidence: AZ 0% vs. NIF 0% vs. TDFL 0% vs. placebo 13%

AZ, acetazolamide; BID, twice daily; DEX, dexamethasone; ESQ, Environmental Symptoms Questionnaire; HAPE, high-altitude pulmonary edema; LLS, Lake Louise scoring system for AMS; N, number of subjects; NIF, nifedipine; NNT, number needed to treat; OD, once daily; PaO_2, partial pressure of oxygen; PASP, pulmonary artery systolic pressure; PCO_2, partial pressure of carbon dioxide; PRED, prednisone; QID, four times daily; RR, respiratory rate; SIL, sildenafil; SR, sustained release; TDFL, tadalafil; TID, three times daily.

TABLE 3. STUDIES OF DEXAMETHASONE FOR RAPID ASCENT ABOVE 3,500 M

Study	*Study design,* N	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Ellsworth et al., 1987)	RCT, 47	AZ 250 mg, DEX 4 mg, or placebo TID 1 day before ascent	Ground vehicle, trekking	Healthy, non-acclimatized lowlander adults on Mt Rainier	4,392	4,392	2,196	AMS Incidence: DEX 25% vs. 76.9% placebo ($p < 0.05$)
(Rock et al., 1987)	RCT, 16	DEX 4 mg every QID vs. placebo 2 days before ascent and during altitude	Airplane, ground vehicle	Healthy, non-acclimatized, lowlander adults on Pikes Peak	4,250	4,300	4,250	AMS Incidence: DEX 31% vs. 60% placebo
(Zell and Goodman, 1988)	RCT, 32	AZ 250 mg BID, DEX 4 mg QID, both or placebo 2 days before ascent and during 2 days at altitude.	Ground vehicle, trekking	Healthy, non-acclimatized adults.	2,710	4,050	1,355	AMS incidence: No significant difference in DEX alone vs. placebo ($p > 0.1$); AZ + DEX 6% vs. placebo 38% ($p = 0.004$)
(Hackett et al., 1988)	RCT, 15	DEX 2 mg QID vs. placebo 1 hour before ascent and during altitude	Helicopter	Healthy non-acclimatized military males	4,400	4,400	4,400	AMS Incidence: 71% DEX vs. 100% placebo had AMS ($p > 0.05$) at 12 hours; AMS Severity (ESQ): DEX 2.6 vs. 4.6 placebo, ($p = 0.09$)
(Rock et al., 1989)	RCT, 28	DEX 0.25 mg, 1 mg or 4 mg BID 12 hrs prior to ascent	Hypobaric chamber	Healthy non-acclimatized, lowlander adults	4,520	4,570	4,520	AMS Incidence: Only 4 mg BID reduced AMS incidence ($p < 0.05$) 1 mg and 0.25 mg BID were ineffective.
(Ellsworth et al., 1991)	RCT, 18	AZ 250 mg TID or DEX 4 mg TID vs. placebo 1 day before ascent	Ground vehicle, trekking	Healthy, non-acclimatized lowlander adults on Mt Rainier	4,392	4,392	2,196	AMS Severity (ESQ): Symptom scores for DEX (0.26 and 0.20) were significantly lower than scores for both AZ (0.80 and 1.20; $P = .025$) and placebo (1.11 and 1.45; $P = .025$).
(Bernhard et al., 1994)	RCT, 23	DEX 4 mg BID or placebo 6 hrs before flight and during 4 days at altitude	Ground vehicle	Healthy, non-acclimatized lowlander adults in Andes	1,645	5,334	1,645	AMS incidence: DEX no difference vs. placebo at 3,689 m; 18% vs. 75% placebo at 5334 m on day 3 ($p = 0.01$); 45% vs. 75% placebo at 5334 m on day 4 ($p = 0.02$).
(Bernhard et al., 1998)	RCT, 13	AZ 500 mg OD + DEX 4 mg BID vs. AZ 500 mg OD + placebo	Airplane, ground vehicle	Healthy, non-acclimatized lowlander adults in Andes. 50% had previously experienced AMS.	1,645	5,334	1,645	AMS Incidence: No significant difference at 3,689 m day 1 and 2; AZ + DEX 0–17% vs. 86–100% AZ + placebo at 5,334 m on Day 3; AZ + DEX 17–50% vs. 43–86% AZ + placebo at 5,334 m on Day 4.
(Hussain et al., 2001)	RCT, 24	AZ 250 mg BID vs. DEX 4 mg BID vs. AZ 250 mg +DEX 4 mg BID vs. placebo 1 day before ascent	Ground vehicle, trekking	Healthy, lowlander, non-acclimatized adults in Karakoram range	4,063	4,578	4,063	AMS severity (ESQ): 5.3 for AZ+ DEX vs. 11.8 placebo vs. 16.8 AZ vs. 14.5 DEX ($p < 0.01$)
(Basu et al., 2002)	RCT, 50	PRED 10 mg vs. PRED 20 mg vs. PRED 40 mg vs. DEX 0·5 mg vs. placebo 2 days before ascent and OD × 3 days at altitude	Helicopter	Healthy non- acclimatized male adults from sea level	3,230	3,450	3,230	AMS Severity (LLS): 1.1 DEX vs. 3.8 placebo at 12 hours ($p < 0.001$); 1.7 DEX vs. 2.1 placebo at 1 day ($p < 0.05$); 3.6 DEX vs. 6.0 placebo at day 2 (<0.001); 2.7 DEX vs. 5.2 placebo at day 3 (<0.001).
(Hussain and Aslam, 2003)	RCT, 44	AZ 250 mg BID vs. DEX 4 mg BID vs. combination of both vs. placebo 1 day before and during ascent.	Ground vehicle, trekking	Healthy adult lowlander non-acclimatized in Karakoram range	4,063	4,578	4,063	AMS Incidence: Compared to placebo and DEX subgroups, AZ significantly increased RR and PaO_2, while decreasing PCO_2 and pH thus expediting acclimatization.

(*continued*)

TABLE 3. (CONTINUED)

Study	*Study design,* N	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Maggiorini et al., 2006)	RCT, 29	TDFL 10 mg vs. DEX 8 mg BID vs. placebo 1 day before ascent and during altitude	Cable car, trekking	HAPE susceptible, non-acclimatized adult mountaineers in Alps	3,459	4,559	3,459	AMS incidence: 30% DEX vs. 89% placebo ($p = 0.02$); HAPE incidence: 0% DEX vs. 78% placebo ($p < 0.001$)
(Fischler et al., 2009)	RCT, 23	DEX 8 mg BID vs. placebo vs. TDFL 10 mg BID 1 day before ascent and during altitude	Cable car, trekking	Non-acclimatized HAPE susceptible adults	3,459	4,559	3,459	Exercise capacity: Compared to placebo, DEX improved O_2 uptake (74% vs. 61%; $p = 0.05$), O_2 kinetics (mean response time 41 seconds vs. 45 seconds; $p = 0.05$) and reduced the ventilatory equivalent for CO_2 (42 vs. 50; $p = 0.01$). During echocardiography at low-intensity exercise DEX lowered pulmonary artery pressure (47 mm Hg vs. 68 mmHg; $p = 0.05$) and improved oxygen saturation (74% vs. 61%; $p = 0.02$)
(Subudhi et al., 2011)	RCT, 20	AZ 250 mg vs. DEX 4 mg vs. Placebo TID 1 day before ascent in chamber and during altitude	Hypobaric chamber	Healthy adults who resided at 1650 m × 1 year	3,225	4,875	3,225	AMS Severity (LLS): AZ 1.3 vs. DEX 1.3 vs. 4.0 placebo ($p < 0.05$)
(Siebenmann et al., 2011)	RCT, 24	DEX 8 mg bid vs. placebo 1 day before ascent	Cable car, trekking	HAPE susceptible, non-acclimatized adults in Alps	3,354	4,559	3,354	AMS Severity (LLS): DEX 3.7 vs. 5.71 placebo ($p = 0.036$); Exercise Capacity: Compared to placebo, DEX decreased workload (Wmax) by 7% ($p = 0.004$), had a 10% smaller decrease in VO_2max ($p = 0.025$), improved resting heart rate ($p = 0.007$) and improved resting SpO_2 (91% vs. 83%; $p = 0.009$)
(Sikri et al., 2019)	RCT, 1,302	Group A: AZ 250 mg SR OD + DEX 4 mg BID; Group B: AZ 250 mg SR OD + placebo 1 day before ascent and during 4 days at altitude. Controls had no medication	Ground vehicle	Healthy Indian Army male soldiers from lowland ethnicity	3,505	3,505	875	AMS Incidence: AZ + DEX 1.56% vs. 1.59% AZ + placebo vs. 19.14% in controls ($p < 0.0001$); AMS Severity (LLS 10–12): AZ + DEX 0% vs. 0% AZ + placebo vs. 4.8% in controls had severe AMS; Exercise capacity: AZ + DEX decreased by 7.83% vs. 12.97% in AZ + placebo
(Chanana et al., 2022)	RCT, 27	DEX 4 mg BID vs. placebo 1 day before flight and during altitude	Airplane	Healthy, non-acclimatized lowlander adults in Himalayas	3,275	3,500	3,275	AMS incidence: DEX 23.1% vs. placebo 21.4% at day 1; 0% vs. 14.3% at day 2; 0% vs. 7.1% at day 3.

VO_2, oxygen consumption per unit time.

TABLE 4. STUDIES OF NIFEDIPINE FOR RAPID ASCENT ABOVE 3,500 M

Study	Study design, N	Intervention	Mode of transport	Population	Delta elevation (m)	Max elevation (m)	Ascent rate (m/day)	Results
(Bartsch et al., 1991)	RCT, 21	NIF 20 mg OD vs. placebo on days 3 and 2 before ascent, then BID the day before ascent, then TID while ascending	Trekking, cable car	HAPE susceptible mountaineers in Alps	3,429	4,559	3,429	HAPE Incidence: NIF 10% vs. 63.6% placebo ($p = 0.01$); AMS Severity: 2.0 vs. 3.9 placebo ($p < 0.01$)
(Hohenhaus et al., 1994)	RCT, 27	NIF 20 mg SR OD vs. placebo on days 3 and 2 before ascent, then BID day 1 prior to ascent, then TID during the ascent	Cable car, trekking	Adult mountaineers with variable susceptibility to AMS, in Alps	3,429	4,559	3,429	AMS incidence: NIF 64.3% vs. 61.5% placebo
(George et al., 2023)	RCT, 120	AZ 250 mg vs. NIF 30 mg SR vs. TDFL 10 mg vs. placebo BID 1 day before ascent and during 3 days at altitude.	Airplane	Military HAPE-susceptible lowlander adults in India who were reintroduced to high altitude 4 weeks after full recovery from HAPE.	3,500	3,500	3,500	HAPE incidence: AZ 0% vs. NIF 0% vs. TDFL 0% vs. placebo 13%

final data were categorized according to the interventions, with 6 articles on oxygen, 29 on acetazolamide (AZ), 17 on dexamethasone (DEX), 3 on nifedipine (NIF), and 5 on phosphodiesterase-5 inhibitors (PDE-5I). Eight articles addressed both AZ and DEX, and one article was included in AZ, NIF, and PDE-5I tables (Fig. 1).

Supplemental oxygen for rapid ascent to VHA

Several studies have documented the effects of hypoxia on cognitive performance and mood following ascent to HA in unacclimatized subjects. These studies consistently found evidence of cognitive impairment, including impaired executive function, working memory, grammatical reasoning, and mood alteration (Crowley et al., 1992; Falla et al., 2024a; Falla et al., 2024b). Field studies on aircraft and helicopter operations at altitudes between 3,000 and 5,500 m indicate that crew members experience symptoms of hypoxia, including cognitive impairment or lightheadedness (Cable, 2003), which are exacerbated by physical activity (Smith, 2005; Smith, 2007; Tristan, 2017). Other studies have also shown a reduction in physical performance at HA, which could negatively affect rescue operations (Clebone et al., 2020; Vogele et al., 2021).

Data on oxygen supplementation at HA are limited and primarily derived from chamber studies involving either normobaric or hypobaric hypoxia. Six studies specifically examined the use of supplemental oxygen during rapid ascent (>300 m/day) to VHA (Table 1). Supplemental oxygen mitigates the negative effects of hypoxia on mood (Legg et al., 2016) and various aspects of cognitive function, including memory, arithmetic, and visual and auditory tasks (Beer et al., 2024; Crowley et al., 1992; Gerard et al., 2000; Legg et al., 2016). In the context of rescue operations, a hypobaric chamber study (Falla et al., 2024b) found that oxygen supplementation improved attention and reaction time in helicopter emergency medical services (HEMS) providers during acute exposure to 4,000 m.

Although supplemental oxygen had a consistent positive effect, additional systemic stressors such as dehydration, sleep deprivation, prolonged shifts, fatigue, and rapid ascent may further exacerbate cognitive impairments at VHA. This is especially important for rescue missions that involve complex decision-making and physical activities.

The positive cognitive effect of supplemental oxygen for pilots at VHA has been acknowledged by aviation agencies in Canada, the United States, Nepal and Europe, and is reflected in their aviation regulations (Table 6). However, these regulations do not specifically comment on oxygen supplementation for rescuers inserted into high-altitude terrain purposely or by accident (grounding). Supplemental oxygen is often reserved only for the pilots and not available for the rescuers in the helicopter.

Exercise capacity is reduced at HA, including physical activities critical to rescue at VHA, like performance of chest compressions (Clebone et al., 2020; Vogele et al., 2021). Simulated and field studies have shown a reduction in chest compression depth within 60–90 seconds of initiating CPR (Vogele et al., 2021), as well as a decrease in the number and depth of effective compressions (Wang et al., 2014). Supplemental oxygen improved CPR performance at a simulated altitude of 2,438 m (Clebone et al., 2020) and improved the 6-minute walk test at 5,050 m in unacclimatized subjects carrying 4 kg weights (Silva-Urra et al., 2011). Although oxygen is unlikely to have a significant impact on rescuer

Table 5. Studies of Phosphodiesterase-5 Inhibitors for Rapid Ascent Above 3,500 m

Study	*Study design,* N	*Intervention*	*Mode of transport*	*Population*	*Delta elevation (m)*	*Max elevation (m)*	*Ascent rate (m/day)*	*Results*
(Richalet et al., 2005)	RCT, 12	SIL 40 mg TID vs. placebo started at altitude	Helicopter	Healthy non-acclimatized, no previous HAPE	4,290	4,350	2,145	HAPE Severity: SIL reduced PASP (−6% vs. sea level) vs. placebo (+21% vs. sea level) ($p < 0.05$); PaO_2 was higher in SIL vs. placebo at rest and exercise. ($p < 0.05$); AMS Incidence: No statistical difference between SIL and placebo
(Maggiorini et al., 2006)	RCT, 29	TDFL 10 mg vs. DEX 8 mg vs. placebo BID 1 day before and during altitude	Cable car, trekking	HAPE susceptible, non-acclimatized adult mountaineers in Alps	3,459	4,559	3,459	AMS incidence: 80% TDFL vs. 89% placebo ($p = 1$); HAPE incidence: 13% TDFL vs. 78% placebo ($p = 0.007$)
(Fischler et al., 2009)	RCT, 23	DEX 8 mg vs. TDFL 10 mg vs. placebo BID 1 day before ascent and during altitude	Cable car, trekking	Non-acclimatized HAPE susceptible adults	3,459	4,559	3,459	Exercise capacity: Compared to placebo, there was no significant difference in TDFL for maximum oxygen uptake (63% predicted vs. placebo 61%), oxygen kinetics (46 seconds mean response time vs. placebo 45 seconds), and ventilatory equivalent for CO_2 (49 vs. placebo 50). However peak oxygen saturation was improved (69% vs. placebo 62%; $p < 0.08$). During echocardiography at low-intensity exercise, TDFL decreased pulmonary artery pressure (57 mmHg vs. placebo 68 mmHg; and had higher oxygen saturation (67% vs. placebo 61%)
(Bates et al., 2011)	RCT, 62	TDFL 50 mg TID vs. placebo	Ground vehicle	Healthy lowlander adults in Andes	1,550	5,200	1,550	HAPE Incidence: No significant difference in PASP with 1 case of HAPE in the placebo group. AMS Severity (LLS): TDFL 83% vs. 63% placebo $p = 0.21$ on Day 2. No significant difference otherwise. AMS Incidence: No significant difference.
(George et al., 2023)	RCT, 120	AZ 250 mg vs. NIF 30 mg SR vs. TDFL 10 mg vs. placebo BID 1 day before ascent and during 3 days at altitude.	Airplane	Military HAPE-susceptible lowlander adults in India who were reintroduced to high altitude 4 weeks after full recovery from HAPE.	3,500	3,500	3,500	HAPE incidence: AZ 0% vs. NIF 0% vs. TDFL 0% vs. placebo 13%

SIL, Sildenafil; TID, three times daily.

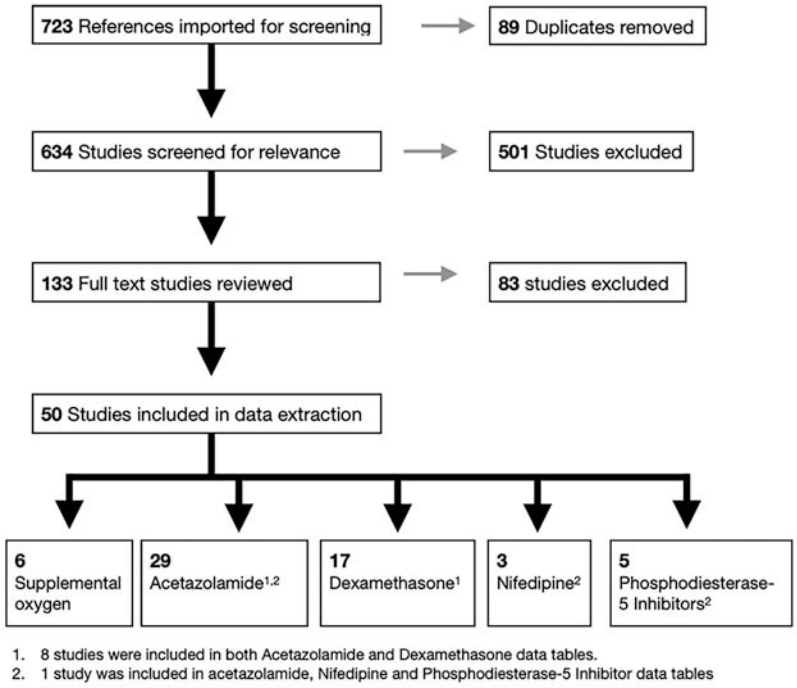


FIG. 1. Literature search results for supplemental oxygen and pharmacologic prophylaxis used in rapid ascent to very high and extreme altitude.

performance for quick patient transport, these studies suggest that supplemental oxygen would help preserve physical performance during rapid ascents above 3,500 m, particularly during activities such as crevasse rescue, avalanche extrication, and human external cargo (HEC) long-line rescue operations.

Supplemental oxygen improved AMS severity scores at 5,050 m in a dose-dependent manner (Silva-Urra et al., 2011). AMS may occur within 6–12 hours of exposure to hypoxia (Bartsch and Swenson, 2013), suggesting that short helicopter trips to VHA place pilots and rescuers at minimal risk of HAI. However, supplemental oxygen may still be required to mitigate the effects of hypoxia on cognitive function and physical performance. When supplemental oxygen is used, there is likely no added benefit from pharmacological prophylaxis; however, this has not yet been studied in the context of rapid ascent to VHA. If the duration at VHA exceeds the oxygen supply, pharmacological prophylaxis should be administered.

Supplemental oxygen can be administered via nasal prongs, simple oxygen masks, or non-rebreather masks, using continuous or on-demand flow systems. At 5,050 m, a supplemental oxygen flow rate of 3 l/min was superior to 1 l/min in reducing AMS symptoms and perceived exertion (Silva-Urra et al., 2011). At simulated altitudes of 8,100 m, flow rates of 1 l/min at rest and 2 l/min during exercise were sufficient to maintain oxygen saturation at a level that

Table 6. Supplemental Oxygen Regional Regulations for Helicopter Pilots and Crew

	Canadian transport agency (Government of Canada, 2023):
Oxygen required for:	Aircraft operators (pilots) and all occupants flying to altitudes of 10,000 ft (3,048 m) to 13,000 ft (3,962 m) for flights longer than 30 minutes Aircraft operators (pilots) and all occupants flying above 13,000 ft (3,962 m) for any duration
	Federal Aviation Agency (USA) (Federal Aviation Administration, 2021):
Oxygen required for:	Aircraft operators (pilots) flying to altitudes of 12,500 ft (3,810 m) to 14,000 ft (4,267 m) for flights longer than 30 minutes. Aircraft operators (pilots) flying to altitudes above 14,000 ft (4,267 m) for any duration Aircraft operators (pilots) and all occupants above 15,000 ft (4,572 m) for any duration
	European Union Aviation Safety Agency (EASA)(European Union Aviation Safety Agency EASA, 2007):
No oxygen required between 13,000 and 16,000 ft (3,962 and4,877 m) if:	Duration <30 minutes, **and** pilot is experienced conducting operations at high altitude without supplemental oxygen, **and** pilot and crew are acclimatized, **and** pilot and crew have received training in hypoxia.
Oxygen required for:	Aircraft operators (pilots) and all occupants flying to altitudes of 13,000 ft (3,962 m) to 16,000 ft (4,877 m) for flights longer than 30 minutes. Aircraft operators (pilots) and all occupants flying above 16,000 ft (4,877 m) for any duration
	Civil Aviation Authority of Nepal (CAAN) (Civil Aviation Authority of Nepal CAAN, 2013):
Oxygen required for:	All crew members and at least 10% of the passengers flying to altitudes of 10,000 ft (3,048 m) to 13,000 ft (3,962 m) for longer than 30 minutes All crew members and passengers flying above 13,000 ft (3,962 m) for any duration.

compensates for hypoxemia-induced symptoms (Wakeham et al., 2023). The effective altitude at rest was lowered by 35% and 66% for 1 l/min and 2 l/min, respectively. With exercise, there was no effect with 1 l/min, but the effective altitude was lowered by 21% with 2 l/min and 47% with 4 l/min (Wakeham et al., 2023). However, the optimal mask, delivery system, flow rate, and utility of oxygen saturation monitoring have not been adequately studied. Lighter battery-powered oxygen concentrators may be a viable alternative to heavy oxygen canisters in rescue operations, particularly for terrestrial rescue teams, providing adequate battery power is available (Sakaue et al., 2008).

Although supplemental oxygen is more accessible for helicopter (air) rescue, the cognitive and physical benefits of supplemental oxygen are still applicable to terrestrial (ground) rescue teams, although its practicality may be compromised due to the weight and limited availability of oxygen canisters. Each terrestrial rescue mission should involve risk analysis to determine the utility of oxygen. Genetically adapted high-altitude rescuers may opt to forgo supplemental oxygen, but many rely on it while guiding above 8,000 m to maintain optimal performance. Consequently, withholding supplemental oxygen from this group should be viewed as altitude-dependent.

Recommendations.

(1) During flight: Pilots and aircraft crews should use supplemental oxygen in accordance with the regional aviation regulations (Table 6). If no regulation exists, supplemental oxygen should be used based on the following guidelines (strong recommendation, moderate-quality evidence):
   - Optional: duration <30 minutes flights below 4,000 m.
   - Recommended: duration >30 minutes flights between 3,500 and 4,000 m.
   - Mandatory: any duration above 4,000 m

(2) Rescuers on the ground after helicopter rapid ascent: Unacclimatized rescuers flown to VHA should use supplemental oxygen on the ground and during longline HEC missions lasting more than 30 minutes at altitudes between 3,500 and 4,000 m and for any duration above 4,000 m (strong recommendation, moderate-quality evidence).

(3) Terrestrial rescuers approaching patients by ground: Terrestrial rescue teams rapidly ascending above 3,500 m should use supplemental oxygen, if it is available and practical (strong recommendation, moderate-quality evidence).

(4) Personnel using supplemental oxygen need not use pharmacological prophylaxis unless the duration of exposure to VHA exceeds the oxygen supply (strong recommendation, low-quality evidence).

Pharmacological Prophylaxis for Rapid Ascent to VHA

Acetazolamide prophylaxis for AMS

AZ enhances altitude acclimatization through carbonic anhydrase inhibition in the kidneys, red blood cells, brain, pulmonary and systemic vasculature, and chemoreceptors (Swenson and Teppema, 2007). Typically, it is administered at a dose of 125 mg every 12 hours, but the optimal dose for rapid ascent to VHA, especially at elevations above 5,000 m, remains unclear. It has been suggested that high-risk groups, such as rescue teams, may benefit from 250 mg every 12 hours although this has not been studied in rapid ascent rescue to VHA (Carlsten et al., 2004; Luks et al., 2024; Toussaint et al., 2021). For teams rapidly ascending to VHA, AZ is discontinued upon descent to the previously acclimatized elevation. However, for those remaining longer at VHA, it should be continued for 2–4 days or until descent, whichever occurs first. (Luks et al., 2024). As AZ has a sulfa moiety, it should not be used in individuals with previous anaphylaxis or Stevens–Johnson Syndrome following sulfonamide administration (Luks et al., 2024).

We identified 29 studies assessing AZ's efficacy in reducing AMS incidence and severity during rapid ascent to VHA. (Table 2). Although the study designs varied, AZ consistently reduced the incidence and severity of AMS compared with placebo after rapid ascent. One study found a nonsignificant trend toward lower AMS incidence when AZ was initiated the night before ascent; however, they also found that day-of-ascent initiation lowered average AMS symptom severity scores (Lipman et al., 2019).

AZ was generally well tolerated, with no significant adverse effects in three studies (Bernhard et al., 1998; Ellsworth et al., 1991; McIntosh et al., 2019; Sikri et al., 2019), although these were not powered for adverse events. Among the reported side effects, paresthesia is the most common (35–91% incidence) (Basnyat et al., 2006; Basnyat et al., 2003; Chow et al., 2005), while taste disturbance, polyuria, nausea, fatigue, and mood change have been reported (Ellsworth et al., 1987; Ke et al., 2013; McIntosh and Prescott, 1986; Zell and Goodman, 1988). Most of these were minor, as only 20% of those who experienced paresthesia were likely to miss a dose (Basnyat et al., 2003).

There are conflicting findings on the cognitive effects of AZ, with studies showing no adverse cognitive effects (Ellsworth et al., 1991), while others have found adverse effects on concentration, cognitive processing speed, reaction time, short-term memory, working memory (Wang et al., 2013), and deterioration in psychological and memory testing (White, 1984).

One smaller study found perception of exercise difficulty was higher with AZ than with placebo at 3,459 m (Bradwell et al., 2014) but a more methodologically sound study (Bradbury et al., 2020) showed no significant effect on exercise performance. If AZ affects exercise capacity, the changes are likely to be small and inconsequential for rescue at VHA.

Given the earliest onset of AMS in rapid ascent was within 5 hours (Hackett et al., 1988) and typically requires 6–12 hours for onset (Bartsch and Swenson, 2013), rapid ascent to VHA for durations less than 3 hours, such as in helicopter rescues, presents minimal risk for AMS. However, if the duration at VHA is expected to exceed 3 hours or is uncertain, pharmacological prophylaxis is recommended if supplemental oxygen is not used.

Recommendations.

(1) AZ should be used for prophylaxis against AMS in rapid ascent rescues to altitudes between 3,500 and 5,000 m if the rescue is expected to last more than 3 hours and immediate deployment is not required (strong recommendation, high quality evidence).

(2) Start AZ the day before ascent if possible. Day-of-ascent initiation is acceptable and confers a benefit over placebo (strong recommendation, moderate-quality evidence).
(3) AZ should be administered at a dose of 250 mg every 12 hours (strong recommendation, moderate-quality evidence).
(4) After rapid ascent, the time spent at VHA is limited to the minimum required to perform the rescue (strong recommendation, low-quality evidence).
(5) AZ should be discontinued upon descent to the previously acclimatized elevation. If prolonged duration is unavoidable at VHA, AZ should be continued during the rescue for 2–4 days or until descent (strong recommendation, low-quality evidence).
(6) AZ should not be used in rescuers with prior anaphylaxis or Stevens–Johnson syndrome caused by sulfonamides (strong recommendation, low-quality evidence).

DEX prophylaxis for AMS

DEX, a corticosteroid, does not facilitate acclimatization to hypobaric hypoxia but is effective in preventing and treating AMS and HACE. It can be used as an alternative to AZ for individuals at moderate to high risk of AMS (Luks et al., 2024) and because of its quicker onset, it is preferable for immediate deployment situations. Typical dosing for AMS prophylaxis is 2 mg every 6 hours or 4 mg every 12 hours, but higher doses of 4 mg every 6 hours may be beneficial in high-risk situations, such as airlifts to altitudes above 3,500 m with immediate physical activity (Hackett et al., 1988; Luks et al., 2024). If rescuers rapidly ascend to VHA and then immediately descend, DEX can be discontinued upon descent to the previously acclimatized elevation. For longer stays at VHA, DEX should be continued for 2–4 days or until descent (Luks et al., 2024). For use for >7 days, DEX must be tapered to avoid adrenal suppression (Luks et al., 2024).

Seventeen studies evaluated the effects of prophylactic DEX on the incidence or severity of AMS during rapid ascent to VHA (Table 3). The studies consistently showed a statistically significant decrease in AMS incidence and severity compared to placebo within the first 24 hours or in the subsequent 2 days at altitude. Abrupt cessation of DEX can unmask AMS symptoms, which may persist for several days (Rock et al., 1987).

Dosing regimens varied among studies, but only 4 mg every 12 hours showed significant improvement in AMS incidence compared to 1 mg and 0.25 mg every 12 hours (Rock et al., 1989).

The combination of AZ and DEX has shown mixed results. In one study, no difference was observed in AMS incidence nor severity at 3,505 m between patients treated with AZ and those treated with both AZ and DEX; however, exercise capacity was better in the combination group (Sikri et al., 2019). Another study found no difference between AZ + placebo and AZ + DEX for the first 2 days after rapid ascent to 3,989 m, although the AZ + DEX group showed improvement in the subsequent 2 days at 5,334 m (Bernhard et al., 1998). Another study found that AZ + DEX significantly decreased AMS incidence compared to placebo, with no significant difference between DEX monotherapy and placebo (Zell and Goodman, 1988). A small, underpowered study found a statistically significant improvement in AMS incidence with AZ + DEX compared to AZ alone, DEX alone, and placebo (Hussain et al., 2001). Although the evidence favoring concurrent use of AZ and DEX is limited, combination therapy may be beneficial for rapid ascent >5,000 m (Bernhard et al., 1998) or missions with prolonged physical ground work at VHA (Sikri et al., 2019), especially if rapid deployment is required.

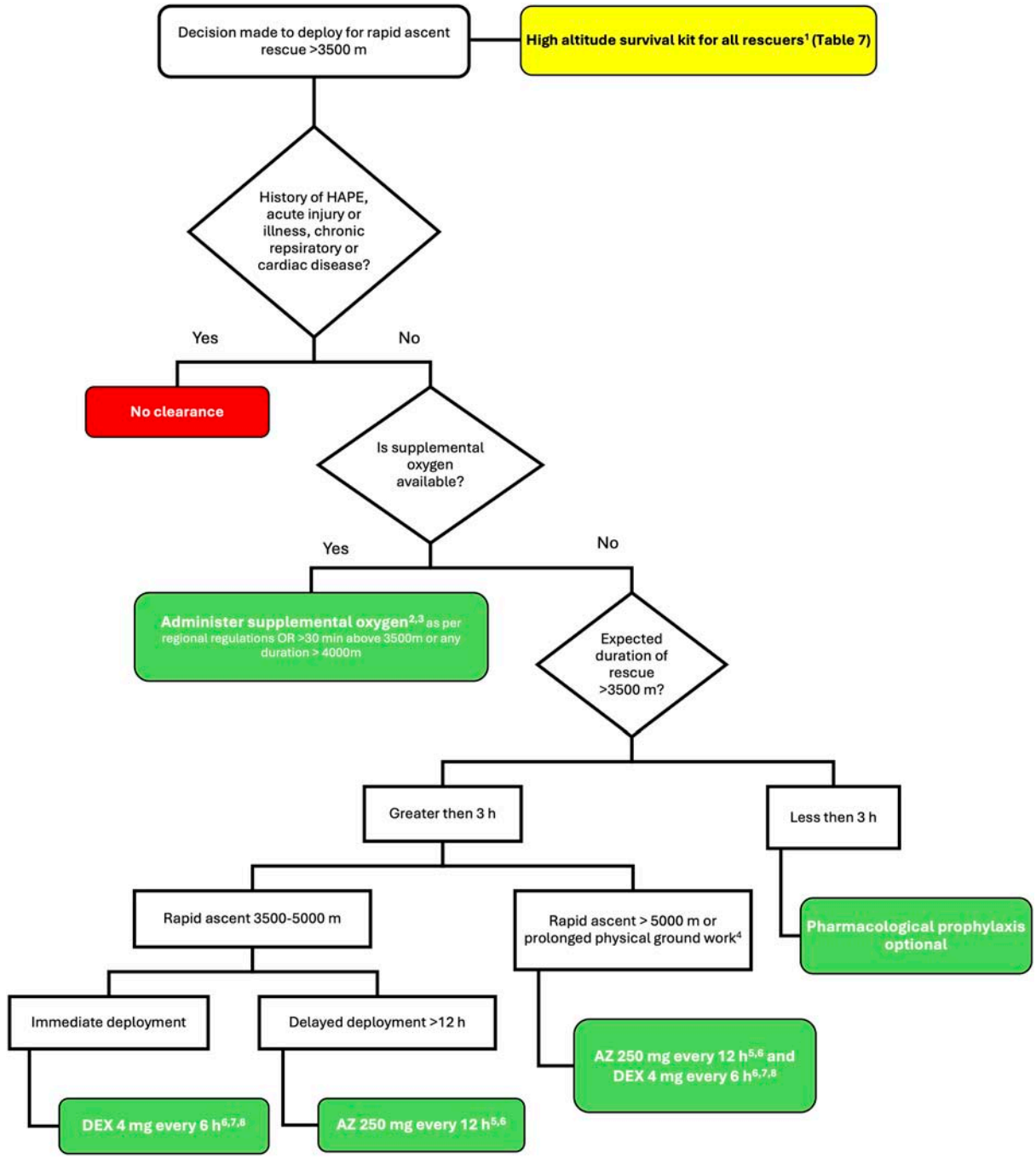
Rare but serious side effects like psychosis have been reported with high dose and long-term use of DEX in hospital settings (Janes et al., 2019), but short-term use was generally well tolerated in our review. Four studies found no adverse effects (Basu et al., 2002; Bernhard et al., 1994; Bernhard et al., 1998; Sikri et al., 2019) and another study found that over 50% of subjects reported fatigue, depression, and insomnia after cessation (Zell and Goodman, 1988). There was no difference in multiple tests of cognitive function between DEX, AZ, and placebo groups at altitude (Ellsworth et al., 1991). DEX has been shown to mitigate the hypoxia-related decline in exercise capacity for HAPE susceptible patients after rapid ascent to HA (Fischler et al., 2009; Siebenmann et al., 2011) especially when combined with AZ (Sikri et al., 2019).

Recommendations.

(1) DEX 4 mg every 6 hours should be used for AMS prophylaxis in rapid ascent rescues when the duration above 3,500 m is expected to exceed 3 hours, and immediate deployment is required (strong recommendation, moderate-quality evidence).
(2) For rescues above 5,000 m with a duration exceeding 3 hours, or missions with prolonged physical ground work, DEX should be used in conjunction with AZ (Strong recommendation, low-quality evidence).
(3) DEX should be used rather than AZ for AMS prophylaxis in individuals with prior anaphylaxis or Stevens–Johnson syndrome from sulfonamides (strong recommendation, low-quality evidence).
(4) DEX should be tapered slowly rather than abruptly stopped if used for more than 7 days (strong recommendation, high-quality evidence).
(5) For prolonged exposure to VHA, DEX should be used for 2–4 days or until descent to the previously acclimatized elevation (strong recommendation, low-quality evidence).

Nifedipine prophylaxis for HAPE

NIF, a calcium channel blocker, is used to prevent and treat HAPE owing to its ability to dilate the pulmonary vasculature. HAPE is unlikely to occur with short VHA exposure, even for those with a history of HAPE, because it typically takes hours or even days to develop. Rescuers with previous HAPE face a higher risk of recurrence with prolonged exposure from a helicopter malfunction or sudden weather changes. Therefore, we recommend excluding individuals with a history of HAPE from rapid ascent rescue to VHA, especially in situations where prolonged exposure is a possibility. If participation is unavoidable for such individuals, supplemental oxygen should be used, or if that is not possible, NIF should be administered at a dose of 30 mg



1. Rescuers and pilots involved in rapid ascent rescue to high altitudes should have an egress plan and survival kit in case of aircraft grounding or unexpected stays at altitude. Prophylactic medications should be started if altitude stay is expected to exceed 3 hours.
2. Supplemental oxygen should be used as per regional aviation regulations for pilots and crew. If no regulations available:
    1. Optional: Duration <30 min flights below 4000 m.
    2. Recommended: Duration >30 min flights between 3500-4000 m.
    3. Mandatory: Any duration above 4000 m
3. Genetically adapted rescuers may choose to abstain from using oxygen.
4. For rescue missions above 5000m or prolonged physical work at very high altitude, recommend Dexamethasone (DEX) 4 mg every 6 h and Acetazolamide (AZ) 250 mg every 12 h due to the increased risk of HACE.
5. Acetazolamide (AZ) for AMS/HACE prophylaxis: 250 mg every 12 h, preferably started 1-2 days prior to ascent.
6. Discontinue after descent to previously acclimatized altitude or if prolonged duration at very high altitude is unavoidable, continue for 2-4 days.
7. Dexamethasone (DEX) for AMS/HACE prophylaxis: 4 mg every 6 hours, started on day of ascent.
8. Taper dosing if Dexamethasone has been taken for more than 5-7 days.

FIG. 2. Supplemental oxygen and pharmacologic prophylaxis algorithm for rescuers and pilots engaging in rapid ascent rescue above 3,500 m.

sustained release (SR) every 12 hours or 20 mg SR every 8 hours. NIF should be continued for 4–7 days if prolonged duration at VHA is necessary, and should be stopped once descent begins (Luks et al., 2024).

Two randomized controlled studies of NIF for HAPE prophylaxis in rapid ascent to HA found statistically significant reductions in HAPE incidence in HAPE-susceptible subjects compared with placebo (Table 4) (Bartsch et al., 1991; George et al., 2023). No study has compared different dosing regimens for NIF.

Recommendations.

(1) Rescuers with a history of HAPE should avoid rapid ascent rescues above 3,500 m unless absolutely necessary (strong recommendation, low-quality evidence).
(2) Rescuers with a history of HAPE who must rapidly deploy to altitudes above 3,500 m should use supplemental oxygen, or if unavailable, initiate NIF 30 mg SR every 12 hours or 20 mg SR every 8 hours (strong recommendation, high-quality evidence).

Phosphodiesterase-5 inhibitors prophylaxis for HAPE

PDE-5I, such as sildenafil and tadalafil dilate pulmonary arterioles and blunt hypoxic pulmonary vasoconstriction at HA (Sylvester et al., 2012). Tadalafil 10 mg every 12 hours is recommended for HAPE prophylaxis in patients with a history of HAPE who cannot take NIF (Luks et al., 2024).

Four randomized controlled trials were identified studying PDE-5I for HAPE prophylaxis in subjects with a history of HAPE during rapid ascent to VHA (Table 5). Tadalafil reduced the incidence of HAPE when compared to placebo (George et al., 2023; Maggiorini et al., 2006) and was equivalent to NIF in only one study (George et al., 2023). No study has shown that PDE-5I is superior to NIF for HAPE prevention in this population.

Recommendations.

(1) Rescuers with a history of HAPE who must rapidly deploy >3,500 m should only use tadalafil (10 mg every 12 hours) if they are unable to take NIF for HAPE prophylaxis (strong recommendation, moderate quality evidence).
(2) NIF and PDE-5I should not be used concurrently (strong recommendation, high quality evidence).

Scope for further research

Future research should focus on applying pharmacologic prophylaxis and supplemental oxygen to rescuers rapidly ascending to VHA in real-world mountain scenarios. Field studies will provide a better understanding of the actual cognitive and physical performance outcomes, incidence, and severity of HAI for different ascent profiles in both ground and air rescue. Research should include comparative dosing of AZ and DEX alone and in combination, various oxygen flow rates and delivery systems, and evaluate the additive effect of oxygen with pharmacologic agents. Further study is necessary to assess the impact of supplemental oxygen, NIF, and PDE-5I on rescuers with a history of HAPE who rapidly ascend to VHA.

Limitations

The lack of appropriately large randomized controlled trials with well-matched study groups and significant differences in study design, including heterogeneity in AMS assessment scoring, make comparisons between studies difficult. Few studies have specifically investigated the rapid ascent of rescuers to VHA, and data are limited on the distinctions between air and ground transportation. Separate recommendations were made for supplemental oxygen and pharmacologic prophylaxis because no studies have examined them together. Although simulated HA environments have been used in previous research, real rescue environments are typically more chaotic, with numerous variables affecting cognitive and physical functioning. There is a lack of research on the optimal method for administering supplemental oxygen and the potential benefits of monitoring pulse oximetry. Future studies should investigate the delta elevation change and not just absolute altitude, as well as the HAI risk mitigation above 5,000 m.

Conclusions

This scoping review provides recommendations for the use of supplemental oxygen and pharmacological prophylaxis to reduce the risk of HAI and improve cognitive and physical performance in unacclimatized rescuers stationed at low altitudes

Table 7. High-Altitude Survival Kit for Rescuers Unable to Descend Following Rapid Ascent >3,500 m

High-altitude survival kit for rescuers and pilots
1 Oxygen supplementation
2 High altitude illness prevention and treatment medications for high-risk ascent: **AMS:** Acetazolamide 250 mg oral every 12 hours **HACE:** Dexamethasone 4 mg every 6 hours (start with 8 mg for treatment) **HAPE:** Nifedipine 30 mg SR oral every 12 hours
3 Personal safety equipment including but not limited to appropriate footwear, adequate clothing, personal shelter, sleeping bag, and climbing equipment (if required) to safely descend if egress plan required or spend the night.
4 Food, hydration, stove, and fuel.
5 Standard first aid kit
6 Consider portable hyperbaric chamber at site or base camp if overnight stay is anticipated

All rescuers and pilots engaging in rapid ascent rescue to very high altitudes should have an egress plan and survival kit in case of aircraft grounding or unexpected stay at altitude. Prophylactic medications should be initiated if the duration at the altitude is expected to be > 3hours.
HACE, high-altitude cerebral edema.

who are swiftly deployed to VHA. We have adapted these recommendations into an algorithm for rescuers rapidly ascending to VHA (Fig. 2) and have included a recommended survival kit as a contingency plan for unexpected prolonged stays at VHA (Table 7). These recommendations will enhance safety and improve the rescue response while serving as a foundation for international guidelines and organization-specific rescue protocols in VHA environments.

Acknowledgment

The authors thank ICAR MedCom for critical discussion of this article.

Authors' Contributions

K.M., S.R., P.P., M.F., and G.S. contributed to the design, implementation of the research, and to the analysis of the results. All the authors contributed to the writing of the article. K.M. and P.P. supervised the project.

Data Availability Statement

Research data supporting this publication are available in Supplementary Data S1, S2 and S3.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

The publication as an open-access article was facilitated by funding from ICAR MedCom.

Supplementary Material

Supplementary Data S1
Supplementary Data S2
Supplementary Data S3

References

Bartsch P, Maggiorini M, Ritter M, et al. Prevention of high-altitude pulmonary edema by nifedipine. N Engl J Med 1991; 325(18):1284–1289; doi: 10.1056/NEJM199110313251805

Bartsch P, Swenson ER. Acute high-altitude illnesses. N Engl J Med 2013;369(17):1666–1667; doi: 10.1056/NEJMc1309747

Basnyat B, Gertsch JH, Holck PS, et al. Acetazolamide 125 mg BD is not significantly different from 375 mg BD in the prevention of acute mountain sickness: The prophylactic acetazolamide dosage comparison for efficacy (PACE) trial. High Alt Med Biol 2006;7(1):17–27; doi: 10.1089/ham.2006.7.17

Basnyat B, Gertsch JH, Johnson EW, et al. Efficacy of low-dose acetazolamide (125 mg BID) for the prophylaxis of acute mountain sickness: A prospective, double-blind, randomized, placebo-controlled trial. High Alt Med Biol 2003;4(1):45–52; doi: 10.1089/152702903321488979

Basu M, Sawhney RC, Kumar S, et al. Glucocorticoids as prophylaxis against acute mountain sickness. Clin Endocrinol (Oxf) 2002;57(6):761–767; doi: 10.1046/j.1365-2265.2002.01664.x

Bates MG, Thompson AA, Baillie JK, et al. Sildenafil citrate for the prevention of high altitude hypoxic pulmonary hypertension: Double blind, randomized, placebo-controlled trial. High Alt Med Biol 2011;12(3):207–214; doi: 10.1089/ham.2011.0007

Beer J, Mojica AJ, Blacker KJ, et al. Relative severity of human performance decrements recorded in rapid vs. gradual decompression. Aerosp Med Hum Perform 2024;95(7):353–366; doi: 10.3357/AMHP.6402.2024

Berger MM, Sareban M, Schiefer LM, et al. Effects of acetazolamide on pulmonary artery pressure and prevention of high-altitude pulmonary edema after rapid active ascent to 4,559 m. J Appl Physiol (1985) 2022;132(6):1361–1369; doi: 10.1152/japplphysiol.00806.2021

Bernhard W, Schalick LM, Gittelsohn A. Dexamethasone for prophylaxis against acute mountain sickness during rapid ascent to 5334 m. Wilderness & Environmental Medicine 1994;5(3):331–338; doi: 10.1580/0953-9859-5.3.331

Bernhard WN, Schalick LM, Delaney PA, et al. Acetazolamide plus low-dose dexamethasone is better than acetazolamide alone to ameliorate symptoms of acute mountain sickness. Aviat Space Environ Med 1998;69(9):883–886.

Bradbury KE, Yurkevicius BR, Mitchell KM, et al. Acetazolamide does not alter endurance exercise performance at 3,500-m altitude. J Appl Physiol (1985) 2020;128(2): 390–396; doi: 10.1152/japplphysiol.00655.2019

Bradwell AR, Myers SD, Beazley M, et al. Exercise limitation of acetazolamide at altitude (3459 m). Wilderness Environ Med 2014;25(3):272–277; doi: 10.1016/j.wem.2014.04.003

Burtscher J, Swenson ER, Hackett PH, et al. Flying to high-altitude destinations: Is the risk of acute mountain sickness greater? J Travel Med 2023;30(4); doi: 10.1093/jtm/taad011

Burtscher M, Bachmann O, Hatzl T, et al. Cardiopulmonary and metabolic responses in healthy elderly humans during a 1-week hiking programme at high altitude. Eur J Appl Physiol 2001;84(5):379–386; doi: 10.1007/s004210100393

Burtscher M, Gatterer H, Faulhaber M, et al. Acetazolamide pre-treatment before ascending to high altitudes: When to start? Int J Clin Exp Med 2014;7(11):4378–4383.

Burtscher M, Hefti U, Hefti JP. High-altitude illnesses: Old stories and new insights into the pathophysiology, treatment and prevention. Sports Med Health Sci 2021;3(2):59–69; doi: 10.1016/j.smhs.2021.04.001

Cable GG. In-flight hypoxia incidents in military aircraft: Causes and implications for training. Aviat Space Environ Med 2003;74(2):169–172.

Carlsten C, Swenson ER, Ruoss S. A dose-response study of acetazolamide for acute mountain sickness prophylaxis in vacationing tourists at 12,000 feet (3630 m). High Alt Med Biol 2004;5(1):33–39; doi: 10.1089/152702904322963672

Castellani JW, Eglin CM, Ikaheimo TM, et al. ACSM expert consensus statement: Injury prevention and exercise performance during cold-weather exercise. Curr Sports Med Rep 2021;20(11):594–607; doi: 10.1249/JSR.0000000000000907

Chanana N, Palmo T, Sharma K, et al. Sexual dimorphism of dexamethasone as a prophylactic treatment in pathologies associated with acute hypobaric hypoxia exposure. Front Pharmacol 2022;13:873867; doi: 10.3389/fphar.2022.873867

Chow T, Browne V, Heileson HL, et al. Ginkgo biloba and acetazolamide prophylaxis for acute mountain sickness: A randomized, placebo-controlled trial. Arch Intern Med 2005; 165(3):296–301; doi: 10.1001/archinte.165.3.296

Civil Aviation Authroity of Nepal CAAN. CAAN flight operations requirements general aviation volume 1 att 2.A-1: Carriage and use of oxygen. In: (Nepal CAAo ed.) 2013.

Clebone A, Reis K, Tung A, et al. Chest compression duration may be improved when rescuers breathe supplemental

oxygen. Aerosp Med Hum Perform 2020;91(12):918–922; doi: 10.3357/AMHP.5698.2020

Crowley JS, Wesensten N, Kamimori G, et al. Effect of high terrestrial altitude and supplemental oxygen on human performance and mood. Aviat Space Environ Med 1992;63(8): 696–701.

Eigenberger P, Faino A, Maltzahn J, et al. A retrospective study of acute mountain sickness on Mt. Kilimanjaro using trekking company data. Aviat Space Environ Med 2014;85(11): 1125–1129; doi: 10.3357/ASEM.4037.2014

Ellsworth AJ, Larson EB, Strickland D. A randomized trial of dexamethasone and acetazolamide for acute mountain sickness prophylaxis. Am J Med 1987;83(6):1024–1030; doi: 10 .1016/0002-9343(87)90937-5

Ellsworth AJ, Meyer EF, Larson EB. Acetazolamide or dexamethasone use versus placebo to prevent acute mountain sickness on Mount Rainier. West J Med 1991;154(3):289–293.

European Union Aviation Safety Agency EASA. Proposed special conditions on auxiliary oxygen system as a supplemental oxygen source applicable to Agusta Westland AW139. In: 2007.

Falla M, Hufner K, Falk M, et al. Simulated acute hypobaric hypoxia effects on cognition in helicopter emergency medical service personnel—A randomized, controlled, single-blind, crossover trial. Hum Factors 2024a;66(2):404–423; doi: 10 .1177/00187208221086407

Falla M, van Veelen MJ, Falk M, et al. Effect of oxygen supplementation on cognitive performance among HEMS providers after acute exposure to altitude: The HEMS II randomized clinical trial. Scand J Trauma Resusc Emerg Med 2024b; 32(1):65; doi: 10.1186/s13049-024-01238-6

Federal Aviation Administration. Oxygen equipment: Use in general aviation operations. In: (Administration FA ed.) 2021.

Fischler M, Maggiorini M, Dorschner L, et al. Dexamethasone but not tadalafil improves exercise capacity in adults prone to high-altitude pulmonary edema. Am J Respir Crit Care Med 2009;180(4):346–352; doi: 10.1164/rccm.200808-1348OC

Fulco CS, Rock PB, Cymerman A. Maximal and submaximal exercise performance at altitude. Aviat Space Environ Med 1998;69(8):793–801.

Gasser B. Stranded because of exhaustion while high-altitude mountaineering in the Swiss Alps: A retrospective nationwide study. Sci Rep 2022;12(1):9011; doi: 10.1038/s41598-022-12917-8

George OS, Singh K, Kumar B, et al. Proof of concept study – alternative pharmacoprophylaxis for high-altitude pulmonary edema: A hospital-based randomized controlled trial. Medical Journal Armed Forces India 2023.

Gerard AB, McElroy MK, Taylor MJ, et al. Six percent oxygen enrichment of room air at simulated 5,000 m altitude improves neuropsychological function. High Alt Med Biol 2000;1(1):51–61; doi: 10.1089/152702900320685

Global Rescue. Rescues in action: Saving high altitude climbers. 2024. Available from: https://www.globalrescue.com/common/blog/detail/high-altitude-mountain-climbing-rescue-experts/

Government of Canada. Canadian aviation regulations (SOR/96-433). Part VI - General operating and flight rules. Subpart 5 - Aircraft regulations. Use of oxygen 605.32. In: (Canada T ed.) 2023.

Greene MK, Kerr AM, McIntosh IB, et al. Acetazolamide in prevention of acute mountain sickness: A double-blind controlled cross-over study. Br Med J (Clin Res Ed) 1981; 283(6295):811–813; doi: 10.1136/bmj.283.6295.811

Grocott MP, Martin DS, Levett DZ, et al. Arterial blood gases and oxygen content in climbers on Mount Everest. N Engl J Med 2009;360(2):140–149; doi: 10.1056/NEJMoa0801581

Guyatt G, Gutterman D, Baumann MH, et al. Grading strength of recommendations and quality of evidence in clinical guidelines: Report from an american college of chest physicians task force. Chest 2006;129(1):174–181; doi: 10.1378/chest.129.1.174

Hackett PH, Rennie D, Levine HD. The incidence, importance, and prophylaxis of acute mountain sickness. Lancet 1976; 2(7996):1149–1155; doi: 10.1016/s0140-6736(76)91677-9

Hackett PH, Roach RC, Wood RA, et al. Dexamethasone for prevention and treatment of acute mountain sickness. Aviat Space Environ Med 1988;59(10):950–954.

Hansen JE, Harris CW, Evans WO. Influence of elevation of origin, rate of ascent and a physical conditioning program on symptoms of acute mountain sickness. Mil Med 1967;132(8): 585–592.

Hillenbrand P, Pahari AK, Soon Y, et al. Prevention of acute mountain sickness by acetazolamide in Nepali porters: A double-blind controlled trial. Wilderness Environ Med 2006; 17(2):87–93; doi: 10.1580/1080-6032-17.2.87

Hohenhaus E, Niroomand F, Goerre S, et al. Nifedipine does not prevent acute mountain sickness. Am J Respir Crit Care Med 1994;150(3):857–860; doi: 10.1164/ajrccm.150.3.8087361

Hussain MM, Aslam M, Khan Z. Acute mountain sickness score and hypoxemia. J Pak Med Assoc 2001;51(5):173–179.

Hussain MM, Aslam M. Hypoxia and pulmonary acclimatisation at 4578 m altitude: The role of acetazolamide and dexamethasone. J Pak Med Assoc 2003;53(10):451–458.

Janes M, Kuster S, Goldson TM, et al. Steroid-induced psychosis. Proc (Bayl Univ Med Cent) 2019;32(4):614–615; doi: 10 .1080/08998280.2019.1629223

Ke T, Wang J, Swenson ER, et al. Effect of acetazolamide and gingko biloba on the human pulmonary vascular response to an acute altitude ascent. High Alt Med Biol 2013;14(2): 162–167; doi: 10.1089/ham.2012.1099

Larson EB, Roach RC, Schoene RB, et al. Acute mountain sickness and acetazolamide. Clinical efficacy and effect on ventilation. JAMA 1982;248(3):328–332.

Legg SJ, Gilbey A, Hill S, et al. Effects of mild hypoxia in aviation on mood and complex cognition. Appl Ergon 2016;53 Pt B:357–363; doi: 10.1016/j.apergo.2015.10.002

Lipman GS, Jurkiewicz C, Winstead-Derlega C, et al. Day of ascent dosing of acetazolamide for prevention of acute mountain sickness. High Alt Med Biol 2019;20(3):271–278; doi: 10.1089/ham.2019.0007

Luks AM, Beidleman BA, Freer L, et al. Wilderness Medical Society clinical practice guidelines for the prevention, diagnosis, and treatment of acute altitude illness: 2024 update. Wilderness Environ Med 2024;35(1_suppl):2S–19S; doi: 10 .1016/j.wem.2023.05.013

Maggiorini M, Brunner-La Rocca HP, Peth S, et al. Both tadalafil and dexamethasone may reduce the incidence of high-altitude pulmonary edema: A randomized trial. Ann Intern Med 2006;145(7):497–506; doi: 10.7326/0003-4819-145-7-200610030-00007

Mallet RT, Burtscher J, Pialoux V, et al. Molecular mechanisms of high-altitude acclimatization. Int J Mol Sci 2023;24(2); doi: 10.3390/ijms24021698

McIntosh IB, Prescott RJ. Acetazolamide in prevention of acute mountain sickness. J Int Med Res 1986;14(5):285–287; doi: 10.1177/030006058601400510

McIntosh SE, Hemphill M, McDevitt MC, et al. Reduced acetazolamide dosing in countering altitude illness: A comparison of 62.5 vs 125 mg (the RADICAL trial). Wilderness Environ Med 2019;30(1):12–21; doi: 10.1016/j.wem.2018.09.002

Milani M, Roveri G, Falla M, et al. Occupational accidents among search and rescue providers during mountain rescue operations and training events. Ann Emerg Med 2023;81(6): 699–705; doi: 10.1016/j.annemergmed.2022.12.015

Ollig P. Rangers respond to multiple high altitude rescues on Denali. In: (Service NP ed.) 2024.

Paal P, Pasquier M, Darocha T, et al. Accidental hypothermia: 2021 update. Int J Environ Res Public Health 2022;19(1); doi: 10.3390/ijerph19010501

Peters MDJ, Marnie C, Tricco AC, et al. Updated methodological guidance for the conduct of scoping reviews. JBI Evid Synth 2020;18(10):2119–2126; doi: 10.11124/JBIES-20-00167

Poudel S, Wagle L, Ghale M, et al. Risk factors associated with high altitude sickness among travelers: A case control study in Himalaya district of Nepal. PLOS Glob Public Health 2025;5(2):e0004241; doi: 10.1371/journal.pgph.0004241

Richalet JP, Gratadour P, Robach P, et al. Sildenafil inhibits altitude-induced hypoxemia and pulmonary hypertension. Am J Respir Crit Care Med 2005;171(3):275–281; doi: 10 .1164/rccm.200406-804OC

Roach RC, Lawley JS, Hackett PH. High altitude physiology. In: Auerbach's Wilderness Medicine. (Auerbach P CT, Harris NS. ed.) Elsevier: 2016; pp. 2–8.

Rock PB, Johnson TS, Cymerman A, et al. Effect of dexamethasone on symptoms of acute mountain sickness at Pikes Peak, Colorado (4,300 m). Aviat Space Environ Med 1987;58(7):668–672.

Rock PB, Johnson TS, Larsen RF, et al. Dexamethasone as prophylaxis for acute mountain sickness. Effect of dose level. Chest 1989;95(3):568–573; doi: 10.1378/chest.95.3.568

Sakaue H, Suto T, Kimura M, et al. Oxygen inhalation using an oxygen concentrator in a low-pressure environment outside of a hospital. Am J Emerg Med 2008;26(9):981–984; doi: 10 .1016/j.ajem.2007.12.009

Salisbury R, H E, Bierling B. The Himalaya by the numbers. A statistical analysis of mountaineering in the Nepal Himalaya, 1950–2019. In: The Himalayan Database: Ann Arbour, Michigan, 2021.

Siebenmann C, Bloch KE, Lundby C, et al. Dexamethasone improves maximal exercise capacity of individuals susceptible to high altitude pulmonary edema at 4559 m. High Alt Med Biol 2011;12(2):169–177; doi: 10.1089/ham.2010.1075

Sikri G, Kotwal A, Singh SP, et al. Is it time to revise the acclimatization schedule at high altitude? Evidence from a field trial in western Himalayas. Med J Armed Forces India 2019; 75(3):251–258; doi: 10.1016/j.mjafi.2018.01.001

Silva-Urra JA, Urizar C, Basualto-Alarcon C, et al. Effects of oxygen supplementation on acute mountain sickness symptoms and functional capacity during a 2-kilometer walk test on Chajnantor plateau (5050 meters, Northern Chile). Wilderness Environ Med 2011;22(3):250–256; doi: 10.1016/j.wem.2011.05.004

Smith A. Hypoxia symptoms reported during helicopter operations below 10,000 ft: A retrospective survey. Aviat Space Environ Med 2005;76(8):794–798.

Smith AM. Acute hypoxia and related symptoms on mild exertion at simulated altitudes below 3048 m. Aviat Space Environ Med 2007;78(10):979–984; doi: 10.3357/asem.1989.2007

Smith B. High altitude rescues - considerations and concerns. Air Med & Rescue 2020:112.

Subudhi AW, Dimmen AC, Julian CG, et al. Effects of acetazolamide and dexamethasone on cerebral hemodynamics in hypoxia. J Appl Physiol (1985) 2011;110(5):1219–1225; doi: 10.1152/japplphysiol.01393.2010

Swenson ER, Teppema LJ. Prevention of acute mountain sickness by acetazolamide: As yet an unfinished story. J Appl Physiol (1985) 2007;102(4):1305–1307; doi: 10.1152/japplphysiol.01407.2006

Sylvester JT, Shimoda LA, Aaronson PI, et al. Hypoxic pulmonary vasoconstriction. Physiol Rev 2012;92(1):367–520; doi: 10.1152/physrev.00041.2010

Toussaint CM, Kenefick RW, Petrassi FA, et al. Altitude, acute mountain sickness, and acetazolamide: Recommendations for rapid ascent. High Alt Med Biol 2021;22(1):5–13; doi: 10 .1089/ham.2019.0123

Tricco AC, Lillie E, Zarin W, et al. PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. Ann Intern Med 2018;169(7):467–473; doi: 10.7326/M18-0850

Tristan LR. Hypoxia occurrence in a military aviator below 3048 m. Aerosp Med Hum Perform 2017;88(1):61–64; doi: 10.3357/AMHP.4722.2017

van Patot MC, Leadbetter G, 3rd, Keyes LE, et al. Prophylactic low-dose acetazolamide reduces the incidence and severity of acute mountain sickness. High Alt Med Biol 2008;9(4): 289–293; doi: 10.1089/ham.2008.1029

Vogele A, van Veelen MJ, Dal Cappello T, et al. Effect of acute exposure to altitude on the quality of chest compression-only cardiopulmonary resuscitation in helicopter emergency medical services personnel: A randomized, controlled, single-blind crossover trial. J Am Heart Assoc 2021;10(23):e021090; doi: 10.1161/JAHA.121.021090

Wakeham D, Tomlinson A, Hackett P, et al. The effects of stepwise reductions in supplemental oxygen on oxygen saturation at rest and during exercise at extreme (simulated) altitude. In: 22nd International Hypoxia Symposium. Lake Louise, Alberta, Canada, 2023.

Wang J, Ke T, Zhang X, et al. Effects of acetazolamide on cognitive performance during high-altitude exposure. Neurotoxicol Teratol 2013;35:28–33; doi: 10.1016/j.ntt.2012.12.003

Wang JC, Tsai SH, Chen YL, et al. The physiological effects and quality of chest compressions during CPR at sea level and high altitude. Am J Emerg Med 2014;32(10):1183–1188; doi: 10.1016/j.ajem.2014.07.007

White AJ. Cognitive impairment of acute mountain sickness and acetazolamide. Aviat Space Environ Med 1984;55(7):598–603.

Willmann G. Ultraviolet keratitis: From the pathophysiological basis to prevention and clinical management. High Alt Med Biol 2015;16(4):277–282; doi: 10.1089/ham.2015.0109

Wu T. Mountain rescue: The highest earthquake in Yushu. High Alt Med Biol 2011;12(1):93–95; doi: 10.1089/ham.2010.1082

Yan X. Cognitive impairments at high altitudes and adaptation. High Alt Med Biol 2014;15(2):141–145; doi: 10.1089/ham .2014.1009

Zell SC, Goodman PH. Acetazolamide and dexamethasone in the prevention of acute mountain sickness. West J Med 1988; 148(5):541–545.

Address correspondence to:
Kyle McLaughlin, MD
Canmore Hospital
1100 Hospital Place
Canmore, AB, T1W 3E6
Canada

E-mail: kyle.mclaughlin@me.com

Received March 8, 2025;
accepted in final form July 12, 2025.